

Summary

To examine the validity of the Western superstition of Friday the 13th as “day of misfortune”, a mathematical model has been developed to evaluate a date’s level of “unluckiness” by using Boltzmann Distribution and Information Theory; furthermore, a prediction model based on Graph Theory that utilizes the Greedy Clique Expansion algorithm is established to predict any specific date’s level of “unluckiness” by considering several of its formal properties.

For initial consideration, both natural disasters and technological disasters are considered. The data covered in the period from 2000 to 2025.

The first main model is the Unluckiness Evaluation Model, divided into two parts: The disaster severity model combines casualty and economic factors into a severity value that measures the disaster’s objective “magnitude”. Then, an expectation model is established to evaluate the subjective impact of each disaster on civilians’ cognition. The main influencing factors that contribute to a civilian’s expectation are: long-term, short-term frequency, and the alarm state. Here, an analogy of the Boltzmann Distribution is used, replacing the “temperature” in a Boltzmann Distribution with the civilian’s “level of expectation”. This yields a “Disaster Severity Expectation Boltzmann Distribution”. Comparing the actual severity with this distribution gives the deviation between the expected and actual severity. After that, Information Theory is used to quantify the level of information, i.e. the “surprisal”, that a disaster’s occurrence brings to a civilian. A higher value of surprisal indicates a greater feeling of “unluckiness”.

By implementing the Unluckiness Evaluation Model, the maximum unluckiness value between 2000 and 2025 is 18.21, the top 25% value is 9.44, and the median is 4.42. The average value of Friday 13th is 5.56. Thus, the results did not show a peak unluckiness value of the Friday 13th days, which disproves the superstition.

The second main part is the Unluckiness Prediction Model. This model first lists several formal characteristics (such as lunar phase) for each disaster; then, all dates and their relationships were transferred into a graph, in which every node represents a date, and the edges denotes whether two dates have highly similar characteristics. Then, the community discovery algorithm Greedy Clique Expansion is implemented to categorize nodes into different cliques where nodes within it share several similar formal characteristics. After that, these cliques can provide a prediction of a new date’s unluckiness value by comparing the new date’s formal characteristics with those of cliques.

Implementing the Unluckiness Prediction Model to the dates in 2025, we found that all of the predicted unluckiness values lie within the interval (7,8), which indicates that there is no significant difference between every day’s unluckiness value. This infers that there is no obvious ‘lucky’ or ‘unlucky’ day, which further disproves the superstition.

Then, the Fourier Amplitude Sensitivity Test (FAST) is conducted to the model, and it quantifies how much output variance is contributed by each input parameter. The results show that short-term frequency is the primary driver of variance; while the severity weights show low first-order effects.

Keywords: “Unluckiness” value, Boltzmann Distribution, Information Theory, Graph Theory, Greedy Clique Expansion, FAST

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1 Introduction

1.1 Background

Throughout human history, certain dates have been culturally stigmatized as "bad" or inherently malevolent. From the Western apprehension toward Friday the 13th to the complex systems of hemerology in ancient Mesopotamia and China, the belief that specific temporal coordinates are predisposed to misfortune is a near-universal human trait. Historically, these beliefs were rooted in myth, astrology, or isolated historical tragedies. However, in a data-rich era, we have both the need and the ability to rigorously investigate whether they are truly "bad" or simply a result of superstition.

1.2 Problem Restatement

To begin with, since it is too complex to fulfill all requirements at once, the major problem has been broken down into various minor tasks that are a lot more approachable. In the end, by combining results from each minor task, it is possible to get a comprehensive solution to the original problem.

Before addressing specific minor tasks, it is also essential to identify key objectives to consider as the model focuses on optimization. The major objectives are as follows:

- **Objective 1:** Statistical Validation of Traditional Beliefs.
- **Objective 2:** Factor Decomposition and Independent Effect Estimation.
- **Objective 3:** Synthesis of a Heterogeneous "Luckiness Metric".
- **Objective 4:** Discovery of "Science-Based Omens" via Graph Theory.
- **Objective 5:** Robustness and Sensitivity Analysis.

With the objectives identified, a specific breakdown of tasks is also provided below.

- **Task 1: Single-Factor Statistical Analysis** Identify a primary "negative" event (e.g., traffic accidents or emergency admissions) and utilize hypothesis testing to determine if Friday the 13th exhibits a statistically significant deviation from the mean frequency of other days.
- **Task 2: Factor Decomposition and Effect Estimation** Decompose the observed frequencies into independent components. This involves using statistical models to estimate the separate contributions of "Friday" (day of week) and "13th" (day of month) to the overall event rate.
- **Task 3: Multi-Criteria Luckiness Metric Construction** Create a comprehensive "Luckiness Index" that integrates heterogeneous events. This task includes justifying the selection of coefficients and weights based on objective data and the "average citizen" perspective.

- **Task 4: Comparative Evaluation and Conclusion** Apply the developed metrics to historical data to conclude whether Friday the 13th is truly an outlier. Furthermore, identify other "good" or "bad" days that may have been overlooked by traditional superstition.
- **Task 5: Science-Based Omen Rule Generation** Formulate a set of predictive rules based on the "formal properties" of a date. This model will move beyond the calendar grid to include astronomical data (lunar phases) and social cycles (proximity to holidays) to predict the "luckiness" of any given day.

1.3 Our Work

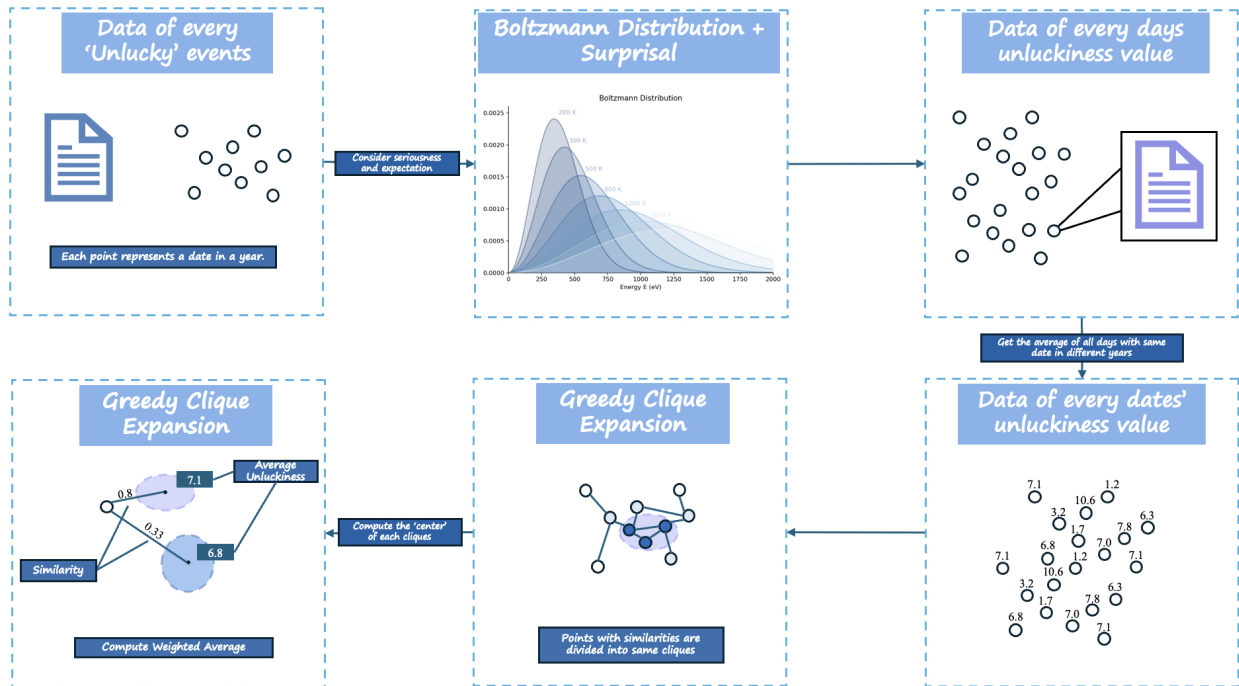


Figure 1: Flowchart of the Model

2 Variables and Parameters

Table 1: Summary of Mathematical Notations

Notation	Variable Name	Description	Unit
M_d	Mortality	Total deaths caused by the disaster	–
I_d	Injury count	Total injuries caused by the disaster	–
A_d	Affected count	Number of civilians affected by the disaster	–
$L_{d,direct}$	Direct loss	Inflation-adjusted direct economic loss	USD

Notation	Variable Name	Description	Unit
$L_{d,\text{indirect}}$	Indirect loss	Inflation-adjusted indirect economic loss	USD
ω_M	Mortality coefficient	Relative weight of mortality in severity	–
ω_I	Injury coefficient	Relative weight of injuries in severity	–
ω_A	Affected coefficient	Relative weight of affected count in severity	–
ω_L	Economic coefficient	Relative weight of direct loss in severity	–
μ	Indirect coefficient	Relative weight of indirect loss in severity	–
$L_{d,i}$	Long-term frequency	Historical disaster frequency at the location	–
$S_{d,i}$	Short-term frequency	Number of days the disaster has persisted	–
$A_{d,i}$	Alarm state	Binary indicator if the disaster was forecast	–
$E_{d,i}$	Expectation value	Degree of civilian expectation for the event	–
$E_{d,i}^*$	Normalized expectation value	Normalized value of civilian expectation	–
D_d	Disaster severity	Calculated severity metric of the event	–
$p(D_d)$	Expected probability	Probability of actual severity in expectations	–
$p^*(D_d)$	Normalized expected probability	Normalized expected probability	–
$I(D_d)$	Information content	Surprisal level caused by the disaster	–
$I^*(D_d)$	Normalized information content	Normalized information content (surprisal)	–
λ_i	Date property	8-dimensional vector of date property	–
F_S	Fitness function	Metric measuring the closeness of a clique	–

3 Assumptions and Justifications

- **Assumption 1: The dataset utilized is comprehensive and free from significant survivorship bias.**

Justification: Given the proliferation of high-resolution remote sensing, satellite imagery, and globalized communication networks in the 21st century, the likelihood of a high-impact disaster remaining unrecorded is negligible. We assume that the reporting threshold for both natural and technological disasters remained consistent across the study period, ensuring that the "unluckiness" value is not skewed by temporal improvements in data collection.

- **Assumption 2: The perceived unluckiness of a given date follows a logarithmic scale relative to the disaster's objective severity.**

Justification: This Weber-Fechner Law posits that the perceived intensity of a stimulus is proportional to the logarithm of its physical intensity. In this context, the psychological "surprisal" is modeled such that a disaster of tenfold severity does not result in a tenfold increase in perceived unluckiness, but rather an additive increment, reflecting the diminishing marginal sensitivity of human response to extreme events.

- **Assumption 3: Intentional attacks and anthropogenic malice do not account for unluckiness.**

Justification: The model focuses exclusively on disasters attributed to force majeure — events resulting from stochastic natural processes or systemic technological failures. We assume that "unluckiness" is a property of random or uncontrollable factors. Consequently, intentional acts (e.g., warfare, terrorism, or targeted criminal activity) are excluded, as these events are driven by human agency and strategic intent rather than the underlying environmental or systemic "background noise" that characterizes the concept of a "bad" day.

- **Assumption 4: Natural disasters tend to be forecast, while technological ones do not.**

Justification: We adopt a heuristic assumption that natural disasters are, to a varying degree, forecasted, whereas technological disasters occur as discrete, unpredictable "Black Swan" events. With modern meteorological and seismic monitoring, the "alarm state value" for natural disasters is non-zero, thereby mitigating the surprisal when the event occurs. Conversely, technological failures are assumed to have an alarm value near zero, as they typically stem from unforeseen human error or latent mechanical fatigue, resulting in a lower expectation coefficient.

- **Assumption 5: The calculation of long-term frequency only regard disasters that happen in the same country.**

Justification: The model assumes that the frequency of a disaster is mediated by geographic and national proximity. Psychological salience is significantly higher for disasters occurring within an individual's own country due to shared infrastructure, national media coverage, and cultural relevance. We assume that events occurring in distant geopolitical regions contribute negligibly to the perceptive frequency of a disaster's occurrence, as the lack of local impact reduces the event to a statistical abstraction rather than a felt experience.

4 Factors

To quantify the contribution of a specific disaster to the aggregate unluckiness value, our model synthesizes two primary analytical dimensions: the objective severity of the event and the subjective expectation of the affected population. As illustrated, the former category evaluates the tangible impact through metrics such as mortality and morbidity rates, the total population affected, and direct economic losses, with each variable weighted via the Analytic Hierarchy Process (AHP) to derive a normalized severity index; simultaneously, the model accounts for the cognitive context of the disaster by calculating civilian expectation — a composite factor derived from the historical frequency of events of the same type within the country, the lead-time provided by predictive forecasts, and the temporal persistence of the catastrophe. Together, these factors define the unluckiness each calendar date possesses in the view of a civilian.

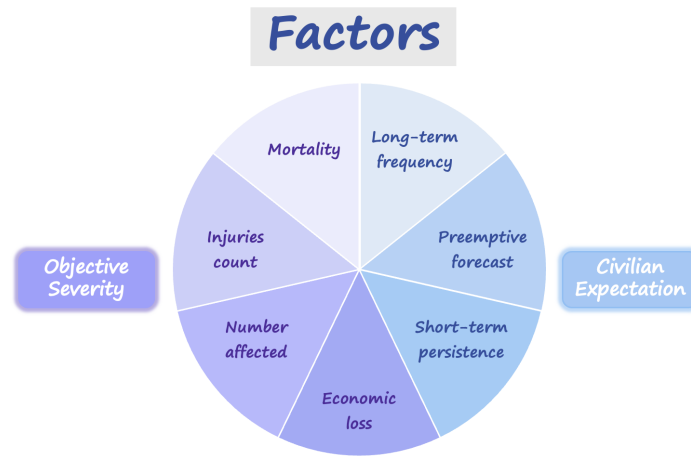


Figure 2: Caption

5 Preliminary Evaluation

To begin with, we chose to focus on a single type of disaster: "Road". We tallied all "Road" events in the database and recorded the frequency at which it is occurring.

Out of all 2211 records of "Road" disasters, only 6 of them occur on Friday the 13th, compared to an average of 10.19 over all possible combinations of a weekday and a day in month. Therefore, according to our statistics, "Friday the 13th" is actually a "good" day.

To determine whether it is possible to decompose the effect into a weekday and a day in the month, we attempt to estimate the frequency based on the weekday and the day in the month. It is obvious to show that the formula is

$$\frac{f_{\text{week}} + f_{\text{month}}}{7 + 31} \quad (1)$$

Using the formula, we estimated the frequency of all possible combinations of a weekday and a day in month, then compared the result of the estimate to real frequency counts. As the control group, we also estimated the frequency by guessing a random number in a certain range.

The estimation based on components resulted in a rooted mean square error (RMSE) of 3.41, compared to 3.78 from random guesswork. Thus, we conclude that there is some relationship between a date and its components, but the relationship is so weak that decomposition cannot serve as a general method of estimation.

6 Unluckiness Evaluation Model

6.1 Model Overview

In order to evaluate a date's level of "luckiness" or "unluckiness", a model that considers the influences of both disaster severity and civilians' expectations is established. The main model consists of two parts: the disaster severity model, which evaluates the objective severity of a disaster based on the mortality and economic loss, and the expectation model, which quantifies the subjective "surprisal" a disaster brings to a civilian by using Boltzmann Distribution and Information Theory. First, the disaster severity value and the "surprisal" value for each disaster are calculated respectively. Then they are multiplied to give the value of "unluckiness" of a disaster. In order to calculate the total "unluckiness" level for a certain day, all disasters that occur on that day are combined logarithmically.

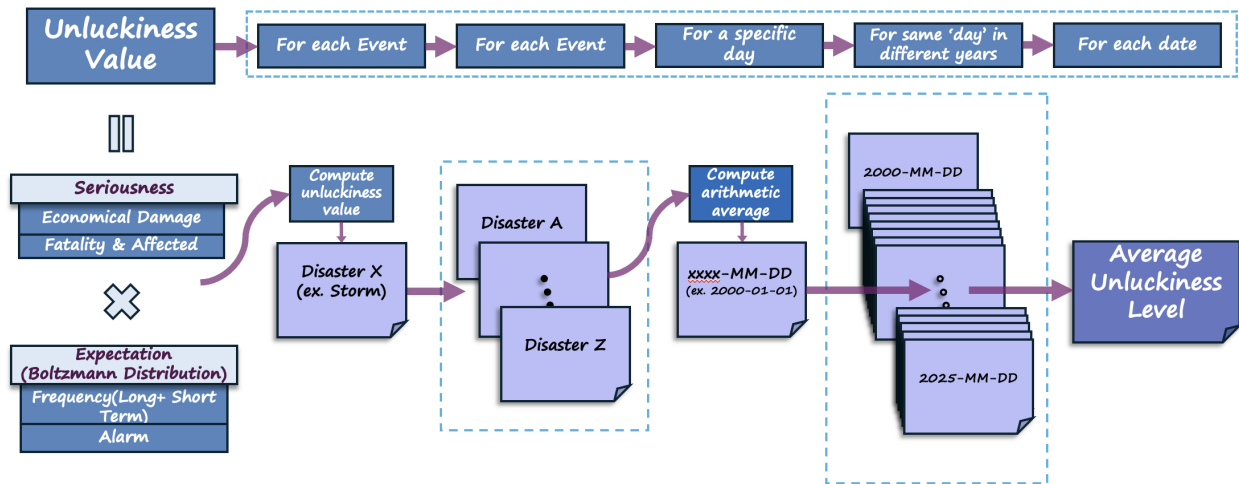


Figure 3: Flowchart for Task 2

6.2 Disaster Severity Model

To evaluate the objective damage a disaster brings to civilians, the disaster severity model is built, which considers several main indicators that manifest a disaster's severity: mortality M_d , number of injured civilians I_d , number of affected civilians A_d , and economic loss L_d , the economic loss is further divided into direct economic loss and indirect economic loss.

The data of mortality, injuries, affected civilians, and direct economic loss were found in EM-DAT database [1], the indirect economic loss is correlated with the number of affected people, given that a larger number of people that are affected by the disaster implies a greater scale of the disaster. The severity value of a disaster is given by:

$$D_d = \omega_M M_d + \omega_I I_d + \omega_A A_d + \omega_L L_{d,\text{direct}} + \mu L_{d,\text{indirect}} \quad (2)$$

By adopting the valuation standards from the World Health Organization Global Burden of Disease Study and using the Disability-Adjusted Life Years conversion method, each indicator's weight is calibrated to represent equivalent lost years of life:

$$\omega_M = 85, \omega_I = 14, \omega_A = 1, \omega_L = 0.1 \quad (3)$$

Also, the indirect economic loss is modeled as a Social Ripple Effect. We define the amplification coefficient μ as a logarithmic function of the affected population, representing the decaying marginal impact of societal shock:

$$\mu = A_d \quad (4)$$

After calculating the exact value of disaster's severity, this value is then normalized using max-min normalization to map its range within $[0, 1]$.

6.3 Expectation Model

6.3.1 Expectation Indicator

The expectation level of an event is referred to the extent to which an event is expected from a civilian's perspective. A higher value of expectation will result in a lower level of surprisal, corresponding to a lower level of "unluckiness", as people generally view the events that are out of their expectations as "unluckier". In this model, the level of expectation is contributed by three main factors: event's long-term frequency, short-term frequency, and whether it is alarmed. Long-term frequency refers to the frequency at which an event occurs within a specific region over an extended period, reflecting the long-term periodicity of the event; short-term frequency refers to the occurrence frequency of the event within a recent period, reflecting the current public expectation regarding its future occurrence. Whether the event is previously alarmed directly determine whether civilians will have an extra preemptive expectation of the event. By combining these factors with Boltzmann Distribution and Information Theory, the surprisal value of a disaster (event) on a specific date can be calculated. The evaluation of each indicator's influence is discussed below.

6.3.2 Long-term Frequency

Long term frequency indicator reflects the influence of long-term periodicity of a disaster on a civilian. It is quantified through the following equation:

$$L_{d,i} = \sum_{t < t_i} 0.99^{-(t_i - t)} \quad (5)$$

In which t_i represents the current time, t represents the times when the disaster happened in the same country, in days, and d represents different types of disasters. This model shows an

exponentially decaying relationship between the time elapsed since a disaster occurs in a country and its influence on civilian's expectation, which is aligned with people's cognitive system. Also, it indicates that a higher frequency of previous disaster occurrence leads to a higher expectation the disaster.

6.3.3 Short-term Frequency

Distinguished from long-term frequency indicator, the short-term frequency indicator measures the impact of a disaster occurring in the recent past on civilian's cognition. For example, if a flood has already persisted for two days, the expectation of it continuing to occur on the next day would be higher. The equation for quantifying its influence is shown below, where Δt represents the time elapsed from the first day of the disaster to the current date.

$$S_{d,i} = \sum_{j=0}^{\Delta t} 1.2^j \quad (6)$$

6.3.4 Alarm State

Alarm state represents whether civilians have a preemptive expectation on the occurrence of the disaster. Its influence is given by:

$$A_{d,i} = \delta_i \quad (7)$$

Where δ_i represents the alarm state of the disaster on date t_i . We assume that natural science usually has a preemptive alarm while disaster caused by technology does not, so that the alarm state is given by:

$$\delta_{d,i} = \begin{cases} 1, & \text{natural disaster} \\ 0, & \text{technological disaster} \end{cases} \quad (8)$$

The total expectation value of a disaster d on date t_i is given by:

$$E_{d,i} = \alpha L_{d,i} + \beta S_{d,i} + \gamma A_{d,i} \quad (9)$$

Where α , β , and γ are the weights for long-term frequency indicator, short-term frequency indicator, and alarm state. By using AHP method, these weights are determined as $\alpha = 1.3$, $\beta = 0.95$, $\gamma = 0.4$. Since the value of $E_{d,i}$ is always a non-negative real number, it can be normalized by using negative exponential transformation to map the raw values onto the interval $[0, 1]$.

$$E_{d,i}^* = 1 - e^{-0.2 \cdot E_{d,i}} \quad (10)$$

6.3.5 Boltzmann Distribution

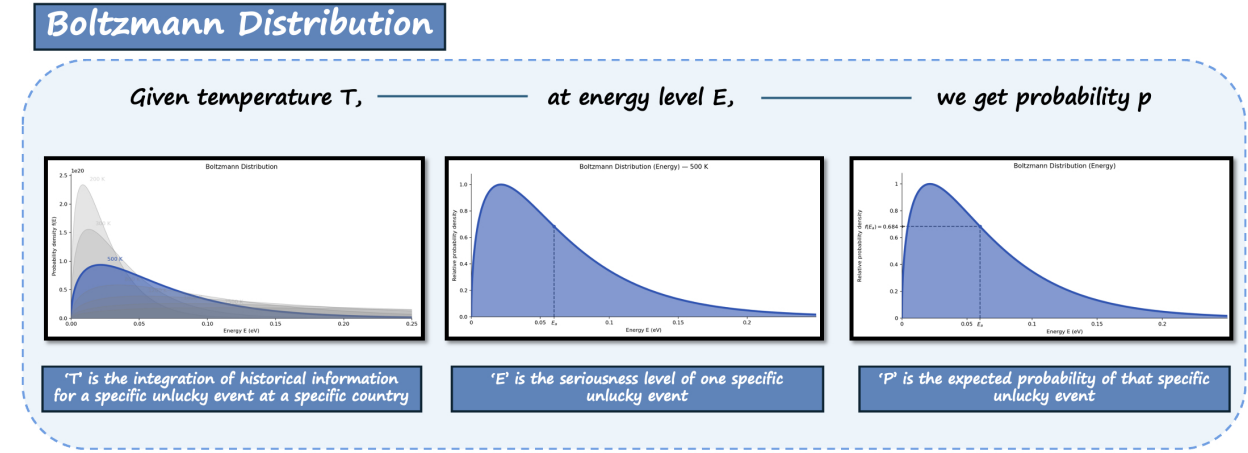


Figure 4: Principle of Boltzmann Distribution

In order to quantify the influence of people’s expectation on the disaster’s “surprisal”, the Boltzmann distribution is used. Boltzmann distribution describes the probability of a system occupying a specific state as a function of that state’s energy and the system’s temperature. Specifically, the probability $p(x)$ of a system at energy state x is expressed as:

$$p(x) = \frac{g(x) e^{-\beta x}}{\int_0^\infty g(s) e^{-\beta s} ds}, \beta = \frac{1}{T^*} \quad (11)$$

It can be seen that the relative probability of a state with “energy” E is proportional to $g(x) e^{-\beta x}$, where $g(x)$ represents microstate multiplicity, which is taken as $g(x) = x^\alpha$ ($\alpha > -1$), the simplest generic representation of multiplicity that enables the model to obtain an analytically solution and clear mechanistic meaning for the shape of the distribution. β is inversely correlated to the system’s “temperature”. The denominator $\int_0^\infty g(s) e^{-\beta s} ds$ represents the “weighted sum” of all possible energy state, ensuring that the total probability of the distribution equals to 1. In order to simplify the expression, the gamma function Γ is introduced:

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt \quad (12)$$

The model of particles in 3-dimension is used, so $\alpha = 0.5$. Substituting the value of α and gamma function $\Gamma(s)$ into the initial distribution function gives:

$$p(x) = \frac{\beta^{1.5}}{\Gamma(1.5)} x^{1.5} e^{-\beta x} \quad (13)$$

This Boltzmann Distribution model can be implemented to measure the deviation of the severity of the disaster from the expected value. First, a predictive model based on Boltzmann distribution is

developed to represent a civilian's expectation of a disaster's severity. In this model, disaster severity levels are treated as "continuous energy states" within the distribution ($D \rightarrow x$). The normalized expected value of disaster $E_{d,i}^*$ serves as the distribution's "temperature", which determine the dispersion of the distribution and the most probable severity perceived by the civilian, i.e. the mode of the distribution ($E_{d,i}^* \rightarrow T^*$). For instance, a civilian with a lower expectation will exhibit an expectation distribution that peaks at a lower severity level, reflecting a subjective belief that small-scale or no disaster is more probable than major disasters. In essence, this "disaster expectation distribution" based on Boltzmann Distribution serves as a quantitative measurement of a civilian's subjective expectation towards a disaster's severity, enabling the model to measure the deviation from actual value to expected value later on.

In order to measure the "level of surprisal" that a disaster brings to a civilian, the probability of the actual disaster severity in one's "disaster expectation distribution" is measured by $p(D)$: a higher probability indicates that the disaster is expected by the civilian, resulting in a lower level of surprisal; lower probability, on the other hand, implies that the disaster is out of one's expectation, leading to a higher level of surprisal. It is important to notice that only the disaster that has a severity larger than the "most probable value" of disaster severity (i.e. the mode) is considered, since disasters that have severity less than the mode value are counted as within one's expectation. The value of the mode in the distribution is calculated by letting its derivative equals to 0:

$$\frac{d}{dx} [g(x) e^{-\beta x}] = 0 \quad (14)$$

So $p_{\text{mode}}(x) = p\left(\frac{0.5}{\beta}\right) = p(0.5 \cdot E_{d,i}^*)$, only the disasters satisfy $D \geq 0.5E_{d,i}^*$ will be considered.

However, noticing that lower value of expectation may cause the low disaster severity region's probability to be larger than 1, it is necessary to conduct a normalization by dividing its maximum value at $D = 0.5E_{d,i}^*$

$$p^*(x) = \frac{p(D)}{p(0.5E_{d,i}^*)} \quad (15)$$

This gives the final "disaster expectation Boltzmann Distribution" for a civilian, where $p^*(x) \in [0, 1]$, the final probability of the actual disaster severity is $p^*(D)$.

6.3.6 Information Theory

After gaining the probability of the actual disaster severity according to a civilian's expectation, information theory is employed to quantify the cognitive impact of the disaster's occurrence. The information content, i.e., the "surprisal" level, of a disaster D is defined as:

$$I(D) = -\log_2(p^*(D)) \quad (16)$$

This indicates that a disaster that has a lower expected probability yields more significant information, thereby enhancing the "surprisal" of human consciousness.

The level of surprisal is then normalized to $(0, 1]$ using logarithmic normalization:

$$I(D)^* = \ln(1 + I(D)) \quad (17)$$

6.4 Total Unluckiness

After gaining a disaster's severity value and its surprisal value, the total effect of a disaster is given by the following equation, where $I(D)^*$ represents the normalized surprisal value and D^* represents the normalized disaster severity.

$$U_d = I(D)^* \cdot D^* \quad (18)$$

According to researches done by Varshney et.al [ah1.1], human perceives the world in a logarithmic way. Thus, a logarithmic function is used to measure the effects of cumulative disasters that happen on the same day. The total unluckiness value of a date is thereby given by:

$$U_{d,t} = \ln(1 + \sum_i^D U_d) \quad (19)$$

Where $U_{d,t}$ means the total disaster's unluckiness value on a certain day, U_d is the value of each disaster's unluckiness value.

6.5 Model Results

The unluckiness evaluation model is implemented to calculate total unluckiness value of a date for each day between 2000.1.1 and 2025.12.31. The maximum value of daily total unluckiness is 18.21, the highest 25% unluckiness value is 9.44, the median value of 4.42, and the last 25% unluckiness is 0. The mean unluckiness of dates that are both 13th in a month and Friday is 5.56, which is slightly higher than the median of all dates. However, its value doesn't exceed the top 25% unluckiness value, indicating that the unluckiness of "13th Friday" lies between the top 25% to top 50% among all dates. Thus, it cannot be said that "13th Friday" is a date with "unluckiness", since its unluckiness value is neither the highest value nor the top 25% value. The Western superstition has been disproven by the model.

In order to further investigate the influence of the date in a year and days of the week on the unluckiness value, we first aggregated the luckiness of the days that have the same date in a year (e.g., 2021.5.4 and 2023.5.4) to gain the total unluckiness value of a specific date (5.4) across multiple years. For each date in a year, the unluckiness values were averaged to neutralize year-to-year variances. The same process was applied to each day of the week to derive the mean unluckiness value associated with different weekdays.

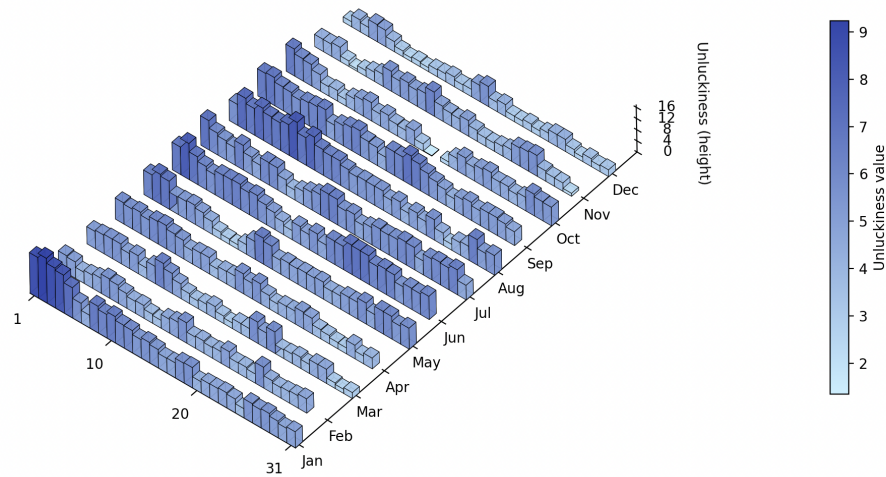


Figure 5: Unluckiness Value for Each Date

Figure 5 shows the average unluckiness value in the date of a year. it can be seen that the first several days in January generally have a greater unluckiness value than other days. This is probably caused by the New Year's Day, since a large amount of people gather together and celebrate on New Year' Day, increasing the possibility of a disaster to happen. Also, August and September have a slightly higher unluckiness value than other months, which we attribute to the effect of hot/cold climate during these two months, since extreme weather enhances the probability of a natural disaster to happen.

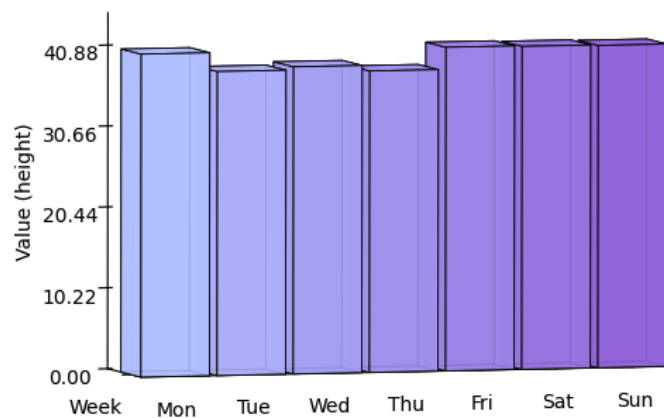


Figure 6: Unluckiness Value for Each Week

Figure 6 displays the average unluckiness value of dates in a week. Tuesday, Wednesday, and Thursday have lower unluckiness than other days in a week. One implication is that they are all weekdays and are not adjacent to weekends, so that less large-scale out-door activities are held on these days.

7 Unluckiness Prediction Model

7.1 Model Overview

Using the **information theory method** established in the previous chapter, it is able to evaluate the unluckiness value for each day from Jan. 1st, 2000, to Dec. 31st, 2025. Subsequently, in order to analyze the underlying scientific causes of these values, multi-dimensional formal characteristics of each date are extracted to create a “formal characteristic set” for each date. By implementing **graph theory**, each date is mapped into a node in the graph, and each weight of the edge represents the similarity of the formal properties between the two dates. The value of “unluckiness” is represented by the weight of each node. Then, the **Greedy Clique Expansion algorithm** is used to divide these nodes into different communities (also called cliques), where nodes within each clique share several common characteristics.

7.2 Characteristics Decomposition

List the formal properties for every date from 2000.1.1 to 2025.12.31, then encode them according to the format in Table 1. For categorical formal properties, use “1” if a date possesses the property and “0” otherwise; for numerical formal properties, use the integer values directly. Subsequently, apply min-max normalization to scale the values of numerical formal properties ("date in a month," "day of the week," and "month") to the range of [0, 1].

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (20)$$

Formal Properties of a Date		
	leap year (LY)	$Z \in [0,1]$
	date in a month (DIM)	$Z \in [1,31]$
	date in a week (DIW)	$Z \in [1,7]$
	month (M)	$Z \in [1,12]$
festival (F)	not festival	0
	before festival	1/3
	during festival	2/3
	after festival	1
lunar phase (LP)	 new moon	0
	 waxing crescent	1/7
	 first quarter	2/7
	 waxing gibbous	3/7
	 full moon	4/7
	 waning gibbous	5/7
	 third/last quarter	6/7
	 waning crescent	1

Table 1: Different Formal Properties of a Date

The data of moon phase and festival are gathered from the Astronomical Applications Department of the U.S. Naval Observatory and the Chinese New Year Institution.

After encoding the formal characteristics, each date will gain a “characteristic set”:

$$\lambda_i = [\text{LY}, \text{DIM}, \text{DIW}, \text{M}, \text{F}, \text{LP}] \quad (21)$$

7.2.1 Calculating similarity level

After giving each date a characteristic set, the similarities of these characteristics between different dates will be calculated using cosine similarity. Now, the characteristic sets λ_i can be seen as vectors in a high-dimensional space. For two nonzero vectors, their cosine similarity is defined as:

$$\text{Cosine Similarity} = \frac{\lambda_A \cdot \lambda_B}{\|\lambda_A\| \|\lambda_B\|} \quad (22)$$

Where $\lambda_A \cdot \lambda_B$ is the dot product of vector λ_A and λ_B , and $\|\lambda_i\|$ represents the norm of a vector. Cosine similarity represents the “angle” between two high-dimensional vectors. It has a range of $[-1, 1]$: a value closer to 1 indicates a higher degree of characteristic similarity, 0 represents irrelevant, while -1 shows a negatively correlated relationship. Iterate through all dates and calculate the cosine similarity between every pair of dates.

7.2.2 Converting Into a Graph

In order to display the dates and their relationships, they are converted into a graph in the following way: set each date as a node, using their corresponding unluckiness value as the node weight. Identify the date pairs that have the top 1% highest cosine similarity value and establish edges between them. The top 1% cosine similarity values fall within the range of $[0.98, 1.0]$, indicating that an adage between two nodes signifies a high degree of similarity.

7.2.3 Greedy Clique Expansion

The greedy clique expansion algorithm is used to divide nodes into different cliques, in which every node shares several similar formal properties. By implementing GCE, it is possible to detect complex overlapping clique structures, as displayed in Figure 7. The main process of greedy clique expansion includes Seed Detection, Clique Expansion, and Overlapping Clique Filtering. Specifically, the algorithm first detects all maximal cliques that have a size of at least k to serve as the initial set of clique (or community) seeds. Subsequently, the algorithm selects seeds sequentially and greedily optimizes a fitness function by iteratively adding neighboring nodes, expanding each seed into a full clique candidate. Finally, by calculating the distance between communities, candidates that are redundant are filtered out, resulting in the final sets of cliques.

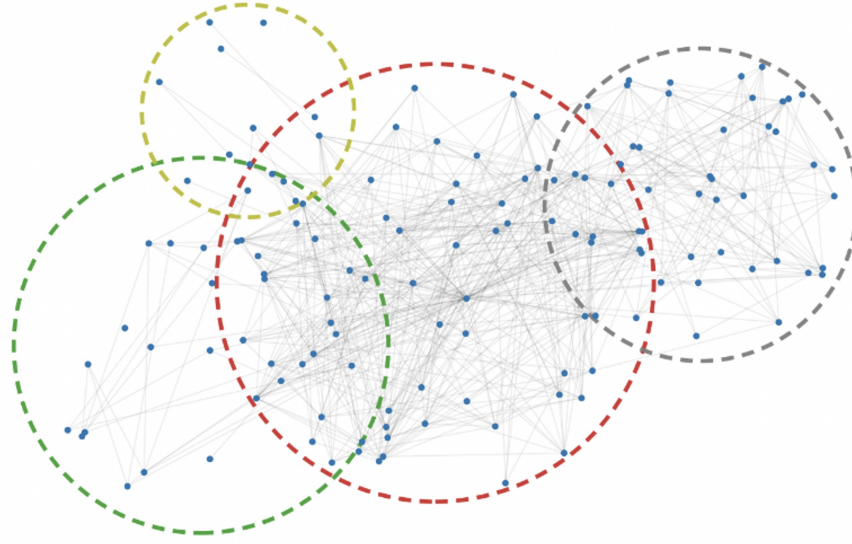


Figure 7: Graph with several distinct cliques, which are outlined with dashed lines

First, the Bron-Kerbosch algorithm is used to choose all maximal cliques, which are fully connected subsets of nodes in a graph that cannot be extended by adding any other adjacent node. To avoid the misidentification of small, coincidentally formed dense structures as clique seeds, a minimum clique size parameter $k = 3$ is established.

The process of clique expansion is driven by a regional fitness function F_S , which assesses the cohesion of a subset (or expanding clique) by evaluating whether adding a node into the clique will enhance or diminish the ratio of internal connection strength.

$$F_S = \frac{k_{\text{in}}^s}{(k_{\text{in}}^s + k_{\text{out}}^s)^\alpha} \quad (23)$$

Where k_{in}^s is the in-degree, or the total number of edges within the clique S , and k_{out}^s denotes the out-degree, or the number of edges connecting nodes inside S to those outside S . α is used to control the size of cliques, which is set to be 1 in this case.

The expansion process will iterate through several steps: from the initial clique seed S , calculate its boundary node set $f(S)$, i.e., the set of all nodes that are connected to at least one node in S but are not contained within S ; then, calculate the fitness function for each boundary node $v \in f(S)$, which is defined as the change in community fitness resulting from adding node v into S , $\Delta F_{S,v} = F_{S \cup \{v\}} - F_S$; traverse through all nodes in $f(S)$ and select the node that has the greatest $\Delta F_{S,v_{\text{max}}}$; if $\Delta F_{S,v_{\text{max}}} > 0$, add node v_{max} into clique S and re-find the boundary node set $f(S)$ and repeat the iteration until no further fitness improvement can be achieved, i.e., $\Delta F_{S,v_{\text{max}}} \leq 0$. The current clique S will be turned into a candidate clique.

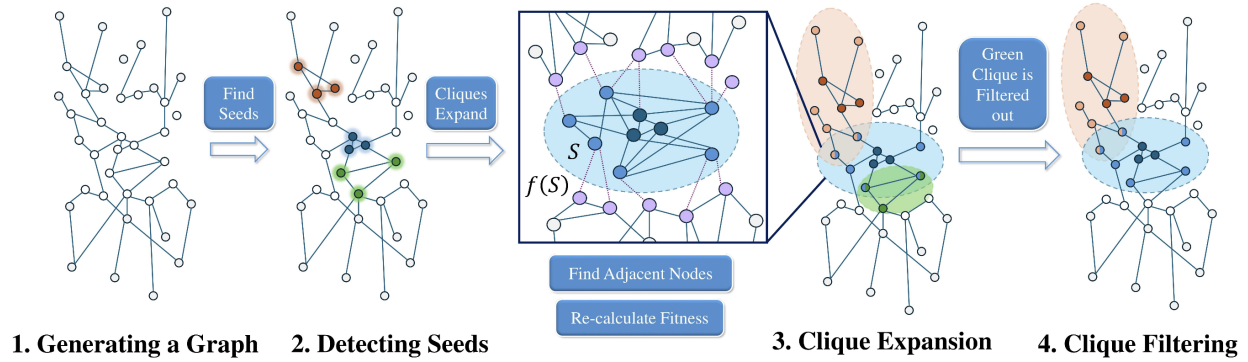


Figure 8: The Four-Stage Pipeline of the Graph-Based Omen Discovery Model

Since multiple seeds may develop identical or highly overlapping candidate clique through the expansion process, redundant clique filtering is performed. By calculating the community distance between pairs of candidate cliques, identical communities are identified, and only the one with the higher fitness value is retained. The community distance between two cliques is calculated by the following equation:

$$\delta_E(S, S') = 1 - \frac{|S \cap S'|}{\min(|S|, |S'|)} \quad (24)$$

Where $S \cap S'$ represents the number of nodes in the intersection of both cliques, i.e., the nodes belonging to both communities, and $\min(|S|, |S'|)$ denotes the number of nodes of the smaller-sized clique. Their ratio quantifies the extent to which the smaller clique is embedded within the larger one. Thus, subtracting this ratio from 1 gives the proportion of the smaller clique that is not covered by the larger one, measuring the non-overlap degree between the two cliques. This value ranged between $[0, 1]$, where a higher value indicates a higher level of distinctiveness. If the clique distance between two cliques is lower than the threshold $\delta_{E, \min}$, then these two cliques are seen to have a high “overlapping degree”, so that only the one that has a higher fitness value will be retained.

7.3 Model Results

We discover that all of the predicted unluckiness value are on the interval (7,8), which indicates that there is no significant difference between every day’s unluckiness value. This may infer that there is no obvious ‘lucky’ or ‘unlucky’ day.

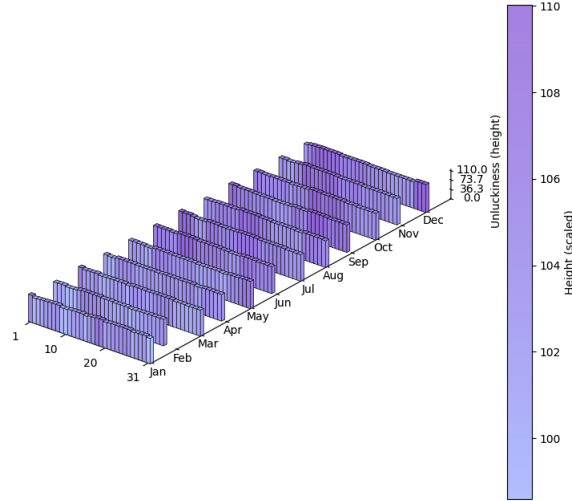


Figure 9: Result of the Unluckiness Prediction Model

8 Sensitivity Analysis

8.1 Theoretical Basis of FAST

The Fourier Amplitude Sensitivity Test (FAST) is a variance-based global sensitivity method. For a model $Y = f(X_1, \dots, X_k)$, the goal is to quantify how much of the output variance $\text{Var}(Y)$ is contributed by each input X_i . The first-order sensitivity index S_i is defined as $S_i = V_i / \text{Var}(Y)$, where V_i is the variance due to X_i alone. FAST estimates these indices by mapping all inputs onto a single scalar sampling variable $s \in [-\pi, \pi]$: each input is written as a periodic function $X_i(s)$ with a distinct frequency ω_i . Running the model along this one-dimensional trajectory produces a sequence $Y(s_j)$ whose Fourier spectrum encodes the influence of each input. The total variance $\text{Var}(Y)$ is recovered from the overall spectral energy, while the contribution V_i is obtained by summing the spectral energy at the frequency band associated with ω_i and its harmonics. In this way, FAST converts the high-dimensional variance decomposition problem into a simple Fourier analysis of $Y(s)$.

8.2 Application of FAST in the Model

To rigorously quantify the uncertainty sources in our Boltzmann-based Omen Model, we employ the **Extended Fourier Amplitude Sensitivity Test (eFAST)**. Unlike local perturbation methods, eFAST performs a global variance decomposition, capable of capturing non-linear interactions within the 7-dimensional parameter space (severity weights $\mathbf{w}$ and expectation scalars λ).

8.2.1 Spectral Variance Decomposition

We assign a distinct characteristic frequency ω_i to each parameter θ_i and oscillate it along a search curve G_i :

$$\theta_i(s) = G_i(\sin(\omega_i s + \varphi_i)), \quad s \in (-\pi, \pi). \quad (25)$$

By evaluating the model output $Y(s)$ along this trajectory, the output variance $\text{Var}(Y)$ can be decomposed via Fourier analysis. The first-order sensitivity index S_i is obtained by summing the spectral power at the characteristic frequency ω_i and its harmonics:

$$S_i = \frac{V_i}{\text{Var}(Y)} = \frac{2 \sum_{p=1}^M (|A_{p\omega_i}|^2 + |B_{p\omega_i}|^2)}{\text{Var}(Y)}, \quad (26)$$

where A_k and B_k are the Fourier coefficients of $Y(s)$ at frequency k , and M is the number of harmonics retained. Here S_i measures the direct (first-order) contribution of θ_i , while the total-effect index S_{Ti} captures the overall influence of θ_i , including its interactions with all other parameters.

8.2.2 Results and Physical Interpretation

The spectral analysis yields two critical insights into the model's physics:

- The scaling of short-term frequency λ_S is the primary driver of variance, with a first-order effect $S_i \approx 0.63$. This aligns with our information-theoretic framework: surprisal is generated primarily by deviations from immediate patterns (short-term anomalies), rather than long-term historical baselines.
- The severity weights $\mathbf{w}$ show low first-order effects ($S_i < 0.05$) but significantly larger total effects ($S_{Ti} > S_i$). The impact of physical damage is not fixed, but is modulated by the system expectation)

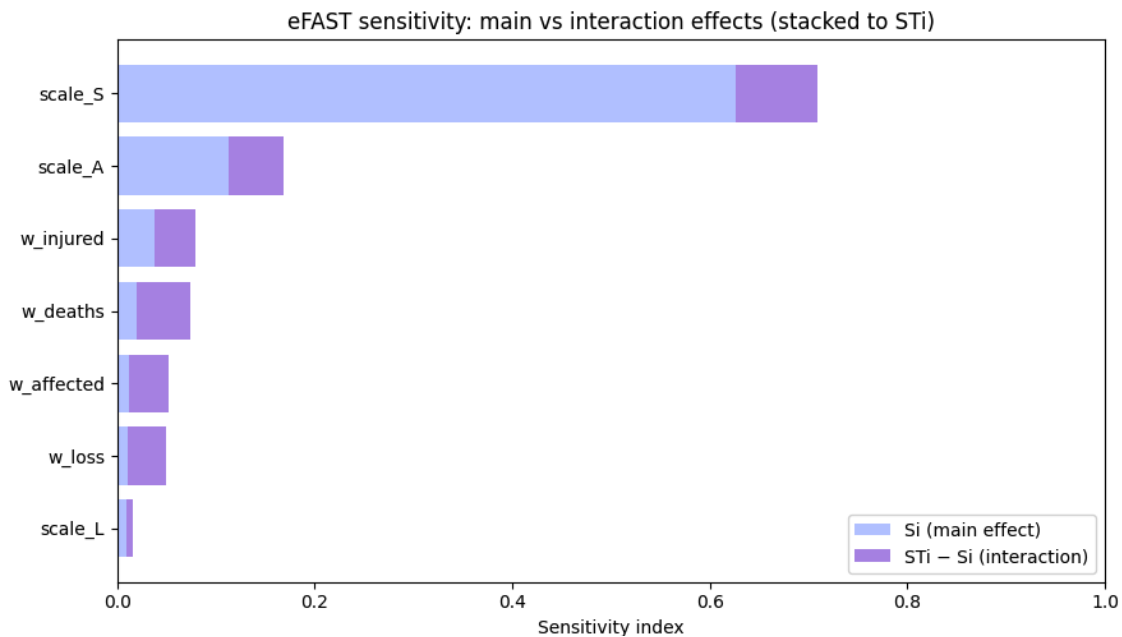


Figure 10: Result of the Sensitivity Analysis

9 Strengths and Weaknesses

Our approach explicitly incorporates psychological expectation into our assessment by representing expectation deviation with an information-theoretic surprisal term. At the same time, we considered the expectation of the residents. Because perceived "luckiness" is subjective rather than purely a function of objective losses. Using surprisal and evaluating expectations allow the metric to capture the value of unluckiness considered by residents subjectively, not only how damaging it is. Moreover, the framework is highly generalizable. It can be extended to longer time range, broader disaster types, and more factors without changing the core structure. A key limitation, however, is the partial dependence on weights, particularly in the seriousness computation. The weights are evaluated by the Analytic Hierarchy Process (AHP) in our referenced article. Since AHP relies on expert judgment and subjective evaluation, these weights may result in bias and decrease in accuracy.

10 Conclusion

The maximum value of daily total unluckiness is 18.21, the highest 25% unluckiness value is 9.44, the median value of 4.42. The mean unluckiness of dates that are both 13th in a month and Friday is 5.56. Its value is not in the top 25% unluckiness value. Thus, it cannot be said that "13th Friday" is a date with "unluckiness", since its unluckiness value is neither the highest value nor the top 25% value. The Western superstition has been disproven by the model.

By implementing the Unluckiness Prediction Model to every dates in 2025, we discovered that all of the predicted unluckiness value are on the interval (7,8), which indicates that there is no significant difference between everyday's unluckiness value. This infers there is no obvious 'lucky' or 'unlucky' day.

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